Forward Kinematics of a Two-Link Planar Arm

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This handout derives the end-effector position of a two-link planar arm from its joint angles. We assume rigid links and revolute joints, and we neglect masses, torques and joint limits.

The elbow position follows from the shoulder angle:
x1 = L1 cos(theta1)
y1 = L1 sin(theta1)

The end-effector position adds the second link, measured relative to the first:
x = L1 cos(theta1) + L2 cos(theta1 + theta2) (1)
y = L1 sin(theta1) + L2 sin(theta1 + theta2) (2)

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Worked example

Our bench arm has L1 = 0.5 m and L2 = 0.3 m.
In the reference pose the shoulder is at theta1 = 30 deg and the elbow at theta2 = 45 deg.

The reachable workspace is an annulus with inner radius |L1 - L2| and outer
radius L1 + L2. With the values above the arm reaches between 0.2 m and 0.8 m.

Limitation: this kinematic model does not account for link deflection or
joint backlash, so measured positions on hardware will differ by a few millimetres.

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